Model Design Document (MDD) – Titanic Data Analysis

1. Introduction

This project focuses on performing Exploratory Data Analysis (EDA) on the Titanic dataset to understand the factors that influenced passenger survival. The objective of this analysis is to explore the dataset, clean and preprocess the data, visualize important variables, and identify relationships between different features and the survival outcome. The insights derived from this analysis help explain which passenger characteristics had the greatest impact on survival during the Titanic disaster.

2. Dataset Description

The dataset used for this analysis is the Titanic passenger dataset. It contains demographic information, ticket details, and survival outcomes of passengers aboard the Titanic.

Dataset Characteristics:

- Total Records: 891 passengers
- Total Columns: 12 features
- Data Types: Numerical and Categorical

Columns in the dataset include:

- PassengerId – Unique identifier
- Survived – Survival status
- Pclass – Passenger class
- Name – Passenger name
- Sex – Gender
- Age – Passenger age
- SibSp – Number of siblings/spouses aboard
- Parch – Number of parents/children aboard
- Ticket – Ticket number
- Fare – Ticket price
- Cabin – Cabin number
- Embarked – Port of embarkation

3. Target Variable

The target variable in this dataset is Survived.

- Survived = 0 → Passenger died
- Survived = 1 → Passenger survived

This variable is used to analyze the survival patterns of passengers based on different attributes.

4. Tools Used

- Python – Programming language for analysis
- Jupyter Notebook – Environment for performing data analysis
- Pandas – Data manipulation and preprocessing
- Matplotlib – Data visualization

- Seaborn – Statistical visualization and correlation heatmaps
- Scikit-learn – Used for Label Encoding

5. Importing Libraries for Data Analysis and Visualization

The following libraries were used in this analysis:

[1]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
```

These libraries provide functionality for data handling, statistical analysis, and visualization.

6. Data Preprocessing

Several preprocessing steps were performed before performing analysis.

[2]:

```
df = pd.read_csv('Titanic-Dataset.csv')
df.head()
```

[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

[3]:

```
df.shape
```

[3]:

```
(891, 12)
```

[4]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

[5]:

```
df.describe()
```

[5]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Handling Missing Values:

[117]:
```
df.isnull().sum()
```

[117]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

- Age column contained missing values and was filled using the median.
- Embarked column missing values were filled using the most frequent value (mode).

[118]:
```
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

df.isnull().sum()
```

[118]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age              0
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         0
dtype: int64
```

Dropping Unnecessary Columns:
- PassengerId: Only a unique identifier.
- Name: High-cardinality text feature.
- Ticket: Mostly unique values.
- Cabin: Contained too many missing values.

[119]:
```
df = df.drop(columns=['PassengerId', 'Name', 'Ticket','Cabin'])
df.head()
```

[119]:

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	male	22.0	1	0	7.2500	S
1	1	1	female	38.0	1	0	71.2833	C
2	1	3	female	26.0	0	0	7.9250	S
3	1	1	female	35.0	1	0	53.1000	S
4	0	3	male	35.0	0	0	8.0500	S

These columns were removed because they do not significantly contribute to the analysis of survival patterns.

Encoding Categorical Variables:

```
[120]:
df = pd.get_dummies(df, columns=['Sex', 'Embarked'], drop_first=True)
df.head()
```

[120…

	Survived	Pclass	Age	SibSp	Parch	Fare	Sex_male	Embarked_Q	Embarked_S
0	0	3	22.0	1	0	7.2500	True	False	True
1	1	1	38.0	1	0	71.2833	False	False	False
2	1	3	26.0	0	0	7.9250	False	False	True
3	1	1	35.0	1	0	53.1000	False	False	True
4	0	3	35.0	0	0	8.0500	True	False	True

Label Encoding Code

```
[121…
le = LabelEncoder()

# Encode Sex column
df['Sex_male']=le.fit_transform(df['Sex_male'])

# Encode Embarked column
df['Embarked_Q']=le.fit_transform(df['Embarked_Q'])
df['Embarked_S']=le.fit_transform(df['Embarked_S'])


df.head()
```

[121]:

	Survived	Pclass	Age	SibSp	Parch	Fare	Sex_male	Embarked_Q	Embarked_S
0	0	3	22.0	1	0	7.2500	1	0	1
1	1	1	38.0	1	0	71.2833	0	0	0
2	1	3	26.0	0	0	7.9250	0	0	1
3	1	1	35.0	1	0	53.1000	0	0	1
4	0	3	35.0	0	0	8.0500	1	0	1

- Sex and Embarked columns were converted into numerical format using encoding to allow correlation analysis.

7. Exploratory Data Analysis (EDA) Steps

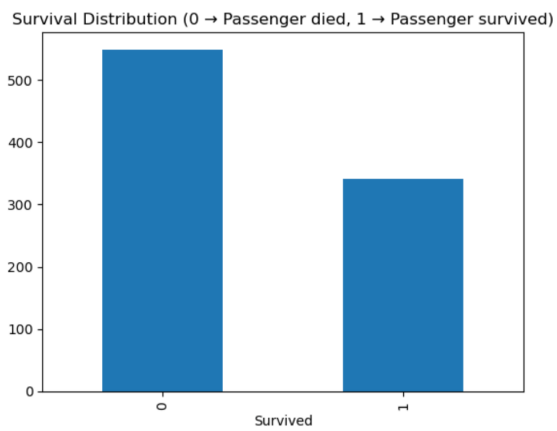
The following steps were performed during EDA:

1. Dataset overview using head(), shape, info(), and describe().
2. Missing value analysis using isnull().sum().
3. Data cleaning and preprocessing.
4. Encoding categorical variables.
5. Univariate analysis of variables such as Age and Survival.
6. Bivariate analysis such as Gender vs Survival and Class vs Survival.
7. Multivariate analysis combining Gender and Passenger Class.
8. Correlation analysis between all numerical variables.
9. Visualization using histograms, bar charts, and correlation heatmaps.

8. Survival Distribution

Survival distribution analysis was performed to understand the overall proportion of passengers who survived and those who did not survive the Titanic disaster. A bar chart was used to visualize the frequency of the survival variable in the dataset. In this dataset, the **Survived column represents the survival status of passengers**, where **0 indicates that the passenger did not survive and 1 indicates that the passenger survived**. From the analysis, it can be observed that the number of passengers who **did not survive is significantly higher than those who survived**. Approximately **62% of the passengers died**, while only about **38% of the passengers survived** the disaster. This indicates that the overall survival rate on the Titanic was relatively low. The survival distribution provides an important baseline understanding of the dataset and highlights that survival was not evenly distributed among passengers. This insight is useful for further analysis to investigate which factors, such as gender, passenger class, age, and fare, may have influenced the chances of survival during the Titanic disaster

[123]:
```
plt.figure()
df['Survived'].value_counts().plot(kind='bar')
plt.title('Survival Distribution (0 → Passenger died, 1 → Passenger survived)')
plt.show()
```

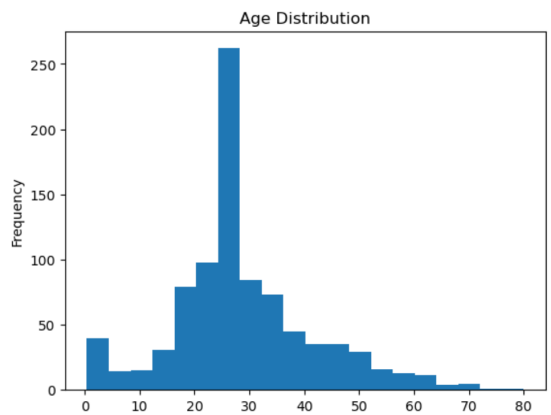


9. Age Distribution

Age distribution analysis was performed to understand how passenger ages are spread across the dataset. A histogram was used to visualize the frequency of passengers within different age ranges. From the analysis, it can be observed that most passengers were between **20 and 40 years old**, indicating that the majority of individuals aboard the Titanic were young to middle-aged adults. There were relatively fewer passengers in the **children category (below 15 years)** and very few passengers in the **elderly category (above 60 years)**. The distribution also shows a **slight right skew**, meaning that while most passengers were adults, there were a few older passengers that extend the distribution toward higher ages. This analysis helps in understanding the demographic composition of the passengers and provides a foundation for further analysis, such as examining whether age had any influence on survival probability during the Titanic disaster.

[124]:

```
plt.figure()
df['Age'].plot(kind='hist', bins=20)
plt.title('Age Distribution')
plt.show()
```



10. Survival by Gender

This analysis compares the survival rates of male and female passengers to understand whether gender influenced survival during the Titanic disaster.

The crosstab() function is used to count how many male and female passengers survived and how many did not survive. The result is then visualized using a bar chart.

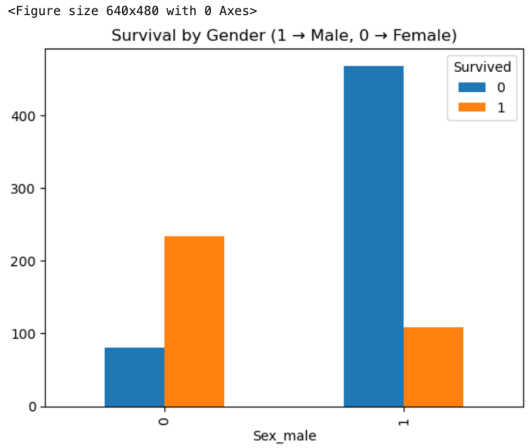
In the encoded dataset:

- **0 represents Female**
- **1 represents Male**

[128]:

```
plt.figure()
pd.crosstab(df['Sex_male'], df['Survived']).plot(kind='bar')
plt.title('Survival by Gender (1 → Male, 0 → Female)')
plt.show()
```

```
<Figure size 640x480 with 0 Axes>
```



11. Survival by Passenger Class

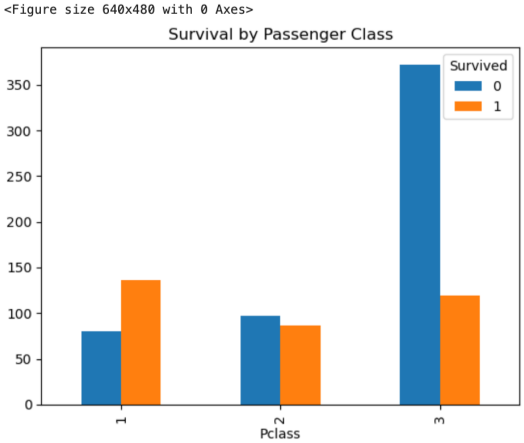
Passenger class analysis was performed to understand how passengers were distributed across different travel classes on the Titanic and how this factor may relate to survival outcomes. The Titanic dataset includes three passenger classes represented by the **Pclass** variable, where **1 indicates first class, 2 indicates second class, and 3 indicates third class**. By analyzing the distribution of passenger classes, it can be observed that the majority of passengers belonged to **third class**, followed by second class and then first class. This indicates that most people traveling on the Titanic were from lower socioeconomic backgrounds who purchased third-class tickets. First-class passengers represented a smaller portion of the dataset but generally had higher ticket fares and better accommodations on the ship. Understanding the distribution of passenger classes is important because it reflects the social and economic hierarchy among passengers and provides useful context for further analysis, particularly when examining how passenger class influenced survival rates during the Titanic disaster.

Sex_male

[129]:

```
plt.figure()
pd.crosstab(df['Pclass'], df['Survived']).plot(kind='bar')
plt.title('Survival by Passenger Class')
plt.show()
```

```
<Figure size 640x480 with 0 Axes>
```



12. Survival by Sex & Passenger Class

Multivariate analysis was performed to examine the combined relationship between multiple variables and the survival outcome. In this case, the analysis focused on understanding how **gender (Sex_male)** and **passenger class (Pclass)** together influenced the survival of passengers. A cross-tabulation was used to compare survival outcomes across different combinations of gender and passenger class. This type of analysis helps identify patterns that may not be visible when analyzing variables individually.

From the analysis, it can be observed that **female passengers in first and second class had the highest survival rates**, while **male passengers, especially those in third class, had the lowest survival rates**. This indicates that both **gender and socioeconomic status (represented by passenger class)** played a significant role in determining survival

during the Titanic disaster. The findings support the historical context that **women and higher-class passengers were given priority during evacuation**. Multivariate analysis therefore provides deeper insight into how multiple factors together influenced survival outcomes rather than examining each variable separately.

10. Multivariate Analysis

[130]:
```
pd.crosstab([df['Sex_male'], df['Pclass']], df['Survived'])
```

[130]:

	Survived	0	1
Sex_male	Pclass		
0	1	3	91
	2	6	70
	3	72	72
1	1	77	45
	2	91	17
	3	300	47

13. Correlation Analysis

Correlation analysis was performed to understand the relationship between different variables in the dataset and how strongly they are related to each other. Correlation measures the strength and direction of the relationship between two numerical variables and is represented by values ranging from **-1 to +1**. A value close to **+1 indicates a strong positive relationship**, meaning that as one variable increases, the other also increases. A value close to **-1 indicates a strong negative relationship**, meaning that as one variable increases, the other decreases. A value close to **0 indicates little or no relationship** between the variables.

In this analysis, a **correlation matrix and heatmap** were used to visualize the relationships between all numerical variables in the dataset. From the results, it can be observed that **gender (Sex_male) has a strong negative correlation with survival**, indicating that male passengers were less likely to survive compared to female passengers. Passenger class (**Pclass**) also shows a negative correlation with survival, meaning passengers in higher classes (lower class number) had better chances of survival. Additionally, **fare shows a positive correlation with survival**, suggesting that passengers who paid higher fares were slightly more likely to survive. Correlation analysis helps identify which variables have the strongest relationships with the target variable and provides useful insights for understanding patterns within the dataset.

[131]:

```
corr = df.corr()
corr.round(2)
```

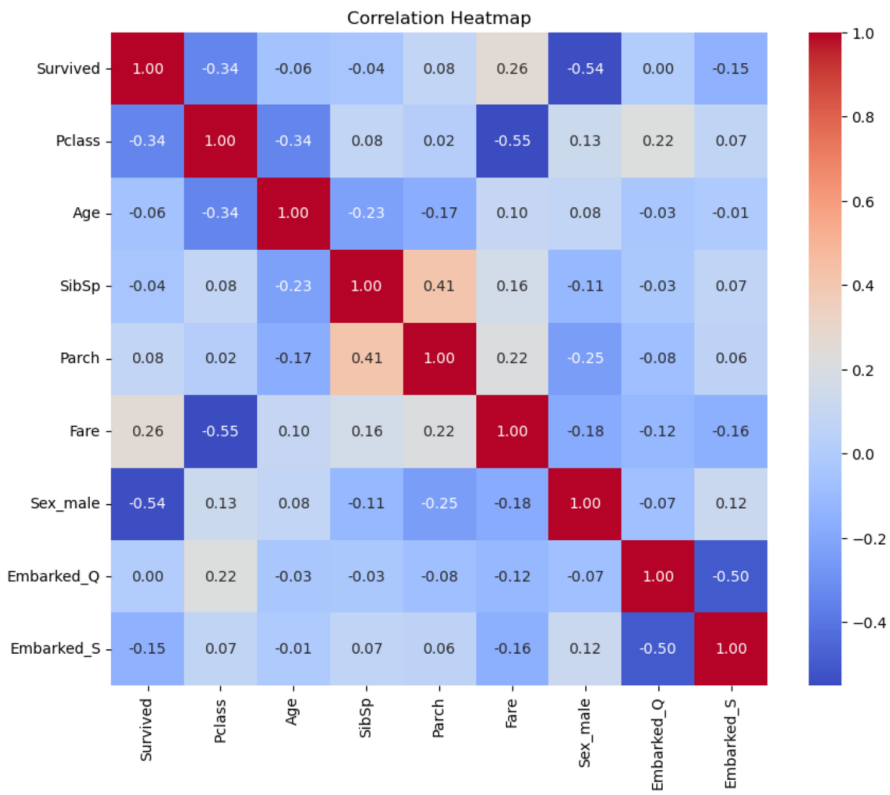
[131]:

	Survived	Pclass	Age	SibSp	Parch	Fare	Sex_male	Embarked_Q	Embarked_S
Survived	1.00	-0.34	-0.06	-0.04	0.08	0.26	-0.54	0.00	-0.15
Pclass	-0.34	1.00	-0.34	0.08	0.02	-0.55	0.13	0.22	0.07
Age	-0.06	-0.34	1.00	-0.23	-0.17	0.10	0.08	-0.03	-0.01
SibSp	-0.04	0.08	-0.23	1.00	0.41	0.16	-0.11	-0.03	0.07
Parch	0.08	0.02	-0.17	0.41	1.00	0.22	-0.25	-0.08	0.06
Fare	0.26	-0.55	0.10	0.16	0.22	1.00	-0.18	-0.12	-0.16
Sex_male	-0.54	0.13	0.08	-0.11	-0.25	-0.18	1.00	-0.07	0.12
Embarked_Q	0.00	0.22	-0.03	-0.03	-0.08	-0.12	-0.07	1.00	-0.50
Embarked_S	-0.15	0.07	-0.01	0.07	0.06	-0.16	0.12	-0.50	1.00

[132…

```
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(),
            cmap='coolwarm',
            annot=True,        # show numbers
            fmt=".2f")         # 2 decimal places

plt.title('Correlation Heatmap')
plt.show()
```



14. Survival Insights from EDA

The analysis reveals several important survival patterns:

- Gender is the strongest predictor of survival.
- Passenger class significantly influenced survival.
- Higher fare passengers had better survival probability.
- Small families had better survival chances than individuals.
- Survival was structured, not random.

15. Conclusion

The exploratory data analysis demonstrates that survival during the Titanic disaster was influenced by several social and demographic factors. Gender and passenger class were the strongest determinants of survival. Female passengers and those traveling in higher classes had significantly better chances of survival. The insights obtained from this analysis highlight how social hierarchy and evacuation priorities influenced survival outcomes during the Titanic disaster.